

ORIGINAL RESEARCH

HEART FAILURE

Pharmacological Treatments in Heart Failure With Mildly Reduced and Preserved Ejection Fraction



Systematic Review and Network Meta-Analysis

Stefanos Zafeiropoulos, MD, MS, PhD,^{a,b} Ioannis T. Farmakis, MD, MS,^{c,d} Ioannis Milioglou, MD, MS,^e Ioannis Doundoulakis, MD, MS, PhD,^{f,g} Eiran Z. Gorodeski, MD,^e Stavros V. Konstantinides, MD, PhD,^{c,h} Lauren Cooper, MD,^{b,i} Stavros Zanos, MD, PhD,^b Stavros Stavrakis, MD, PhD,^j Grigorios Giamouzis, MD, MS, PhD,^{k,l} Javed Butler, MD, MPH, MBA,^{m,n} George Giannakoulas, MD, PhD^d

ABSTRACT

BACKGROUND Medical treatment for heart failure with preserved ejection (HFpEF) and heart failure with mildly reduced ejection fraction (HFmrEF) has weaker evidence compared with reduced ejection fraction, despite recent trials with an angiotensin receptor neprilysin inhibitor (ARNI) and sodium glucose co-transporter 2 inhibitors (SGLT2is).

OBJECTIVES The authors aimed to estimate the aggregate therapeutic benefit of drugs for HFmrEF and HFpEF.

METHODS The authors performed a systematic review of MEDLINE, CENTRAL, and Web of Science for randomized trials including patients with heart failure (HF) and left ventricular ejection fraction (LVEF) >40%, treated with angiotensin-converting enzyme inhibitors or angiotensin receptor blockers (analyzed together as renin-angiotensin system inhibitors [RASi]), beta-blockers (BBs), mineralocorticoid receptor antagonists (MRAs), digoxin, ARNI, and SGLT2i. An additive component network meta-analysis was performed. The primary outcome was a composite of cardiovascular (CV) death and first hospitalization for heart failure (HHF); secondary outcomes were CV death, total HHF, and all-cause mortality.

RESULTS The authors identified 13 studies with a total of 29,875 patients and a mean LVEF of $56.3\% \pm 8.7\%$. ARNI, MRA, and SGLT2i separately, but not RASi, BB, or digoxin, reduced the primary composite outcome compared with placebo. The combination of ARNI, BB, MRA, and SGLT2i was the most effective (HR: 0.47 [95% CI: 0.31-0.70]); this was largely explained by the triple combination of ARNI, MRA, and SGLT2i (HR: 0.56 [95% CI 0.43-0.71]). Results were similar for CV death (HR: 0.63 [95% CI 0.43-0.91] for ARNI, MRA, and SGLT2i) or total HHF (HR: 0.49 [95% CI 0.33-0.71] for ARNI, MRA, and SGLT2i) alone. In a subgroup analysis, only SGLT2i had a consistent benefit among all LVEF subgroups, whereas the triple combination had the greatest benefit in HFmrEF, robust benefit in patients with LVEF 50% to 59%, and a statistically marginal benefit in patients with LVEF $\geq 60\%$.

CONCLUSIONS In patients with HF and LVEF >40%, the quadruple combination of ARNI, BB, MRA, and SGLT2i provides the largest reduction in the risk of CV death and HHF; driven by the robust effect of the triple combination of ARNI, MRA, and SGLT2i. The benefit was more pronounced in HFmrEF patients. (J Am Coll Cardiol HF 2024;12:616-627)
© 2024 by the American College of Cardiology Foundation.

From the ^aElmezzi Graduate School of Molecular Medicine, Northwell Health, Manhasset, New York, USA; ^bFeinstein Institutes for Medical Research at Northwell Health, Manhasset, New York, USA; ^cCenter for Thrombosis and Hemostasis, University Medical Center Mainz, Germany; ^dDepartment of Cardiology, AHEPA University Hospital, Thessaloniki, Greece; ^eHarrington Heart and Vascular Institute, University Hospitals, Cleveland, Ohio, USA; ^fAthens Heart Center, Athens Medical Center, Athens, Greece; ^gFirst Department of Cardiology, National and Kapodistrian University, "Hippokraton" Hospital, Athens, Greece; ^hDepartment of Cardiology, Democritus University of Thrace, Alexandroupoli, Greece; ⁱDepartment of Cardiology, Donald and Barbara Zucker

Heat failure with preserved ejection fraction (HFpEF) (left ventricular ejection fraction [LVEF] $\geq 50\%$) accounts for up to one-half of all heart failure (HF) patients.¹ In contrast to heart failure with reduced ejection fraction (HFrEF) where 4 effective drug classes compose the guideline-directed medical therapy (GDMT), there is no such comprehensive GDMT for HFpEF. Specifically, beyond sodium glucose co-transporter 2 inhibitors (SGLT2is), which recently proved to be effective, 2 two additional drug classes, mineral receptor antagonists (MRAs) and angiotensin receptor blockers or angiotensin receptor-neprilysin inhibitors (ARNI), are modestly recommended by the American Heart Association (AHA)/American College of Cardiology (ACC)/Heart Failure Society of America (HFSA) guidelines.²

Historically, drugs tested in these patients led to overall disappointing results with only marginal reductions in hospitalizations for heart failure (HHF). Recently, empagliflozin and dapagliflozin met their primary endpoint in trials of heart failure with mildly reduced ejection fraction (HFmrEF)/HFpEF,^{3,4} whereas sacubitril/valsartan marginally missed statistical significance for its primary endpoint.⁵ The results of these trials cannot determine the most effective pharmacological combinations. Network meta-analyses allow for estimations and comparisons of treatment effects between different pharmacological combinations if therapies have an additive effect. We conducted a systematic review and network meta-analysis to estimate and compare the additive effects of drug therapies for HFmrEF and HFpEF.

SEE PAGE 628

METHODS

This systematic review and network meta-analysis was conducted according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) extension statement for Network Meta-Analyses.⁶ This study's protocol was prospectively registered in the Open Science Framework ([10.17605/OSF.IO/HC9SK](https://doi.org/10.17605/OSF.IO/HC9SK)). All analyses were based on

previously published studies, and thus, neither ethical approval nor informed consent from patients was needed.

SEARCH STRATEGY AND SELECTION CRITERIA.

We systematically searched MEDLINE, CENTRAL (Cochrane Central Register of Controlled Trials), and Web of Science databases from inception to September 2022 to identify eligible randomized controlled trials (RCTs). We created a search string for MEDLINE and modified it accordingly for the other databases by using relevant terms. The detailed search strategy is shown in the [Supplemental Appendix](#).

We considered RCTs comparing the efficacy of the main drug classes used in HFmrEF or HFpEF. Eligible drug classes included angiotensin-converting enzyme inhibitors, angiotensin receptor blockers—analyzed together as renin-angiotensin system inhibitors (RASi)—ARNI, beta-blockers (BBs), MRAs, SGLT2i, vericiguat, and digoxin. Eligible trials enrolled adult patients with HF and LVEF $>40\%$. Studies in acute HF or without prespecified endpoint adjudication were excluded. The primary outcome was a composite of time to cardiovascular (CV) death and first HHF, whereas the secondary outcomes included the CV death and total HHF, as well as all-cause mortality.

All search results were imported to Rayyan following the duplicates' removal. Two independent investigators (S.Z., I.T.F.) screened titles, abstracts, and full-text articles. Discrepancies were resolved by consensus at each step with a third independent investigator (I.M.).

STRATEGY FOR DATA SYNTHESIS. We considered patients on each background therapy if at least 50% of each trial's population were on the specific drug class at baseline, as previously described.⁷ Substudies related to the parent trials were retrieved to extract data from subgroups that were treated entirely with the aforementioned drug classes.

ABBREVIATIONS AND ACRONYMS

ARNI = angiotensin receptor-neprilysin inhibitor

BB = beta-blocker

CV = cardiovascular

GDMT = guideline-directed medical therapy

HFmrEF = heart failure with mildly reduced ejection fraction

HFpEF = heart failure with preserved ejection fraction

HFrEF = heart failure with reduced ejection fraction

HHF = hospitalizations for heart failure

LVEF = left ventricular ejection fraction

MRA = mineral receptor antagonist

RASI = renin-angiotensin system inhibitor

RCT = randomized controlled trial

SGLT2i = sodium glucose co-transporter 2 inhibitor

School of Medicine at Hofstra/Northwell, North Shore University Hospital, Manhasset, New York, USA; ¹Heart Rhythm Institute, University of Oklahoma Health Science Center, Oklahoma City, Oklahoma, USA; ²Department of Cardiology, University of Thessaly, Larissa, Greece; ³Faculty of Medicine, School of Health Sciences, University of Thessaly, Greece; ⁴Department of Medicine, University of Mississippi School of Medicine, Jackson, Mississippi, USA; and the ⁵Baylor Scott and White Research Institute, Dallas, Texas, USA.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

We used a frequentist framework to conduct a random-effects network meta-analysis and produce direct and indirect effect estimates of interventions. The effect estimate was the HR with the 95% CI for the time-to-event analysis. We created a connected network of the different treatment combinations. We evaluated comparisons between the different treatment combinations, as well as between the individual treatment components in an additive network meta-analysis model. We assumed that the effect of treatment combinations is equal to the sum of the effect of each component in the additive model, and therefore, the common components on both sides of a comparison cancel out. For example, a comparison of A + B vs B gives the (absolute) effect of A alone (vs placebo). We tested the additivity assumption by evaluating the heterogeneity difference between the standard and the additive model. First, we calculated direct and indirect effect estimates for all different treatment combinations based on the background therapy within a trial as well as for the individual components present in the network. We then further calculated the effect estimates for all possible combinations between ARNI (or RASi), MRA, and SGLT2i. Subgroup analyses of patients with HFmrEF and HFpEF as well as patients with HFpEF and LVEF 50% to 59%, and $\geq 60\%$ were performed. We used the TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist) Americas treatment estimates in our primary analysis and the full TOPCAT treatment estimates in a sensitivity analysis.

QUALITY OF EVIDENCE. We used the CINeMA (Confidence in Network Meta-analysis) framework to evaluate the confidence in the network meta-analysis results. CINeMA is a network meta-analysis version of the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) method that considers 6 domains: 1) within study bias; 2) reporting bias; 3) indirectness; 4) imprecision; 5) heterogeneity; and 6) incoherence.⁸ We used the ROB-2 tool to assess the within-study bias across 5 domains, that is, randomization process, deviations from intended interventions, missing outcome data, measurement of outcome, and selection of the reported result.⁹ We tested reporting/publication bias with a visual assessment of funnel plots and the Egger's test. Indirectness, which captures transitivity in the network, was assessed as low. Imprecision compares the treatment effects included in the 95% CI with the range of equivalence using a cutoff of OR: 2.0 as clinically important effect size. We used both global approaches, such as the Cochran's Q and the I^2 statistic (25% low, 25% to 50% moderate, and $>50\%$ high

heterogeneity), and local ones, such as analyzing consistency across direct and indirect comparisons with the node-splitting method to measure consistency. The extent of certainty that a treatment is better than another treatment was quantified with the P-score measured on a scale from 0 (worst ranking) to 1 (best ranking).

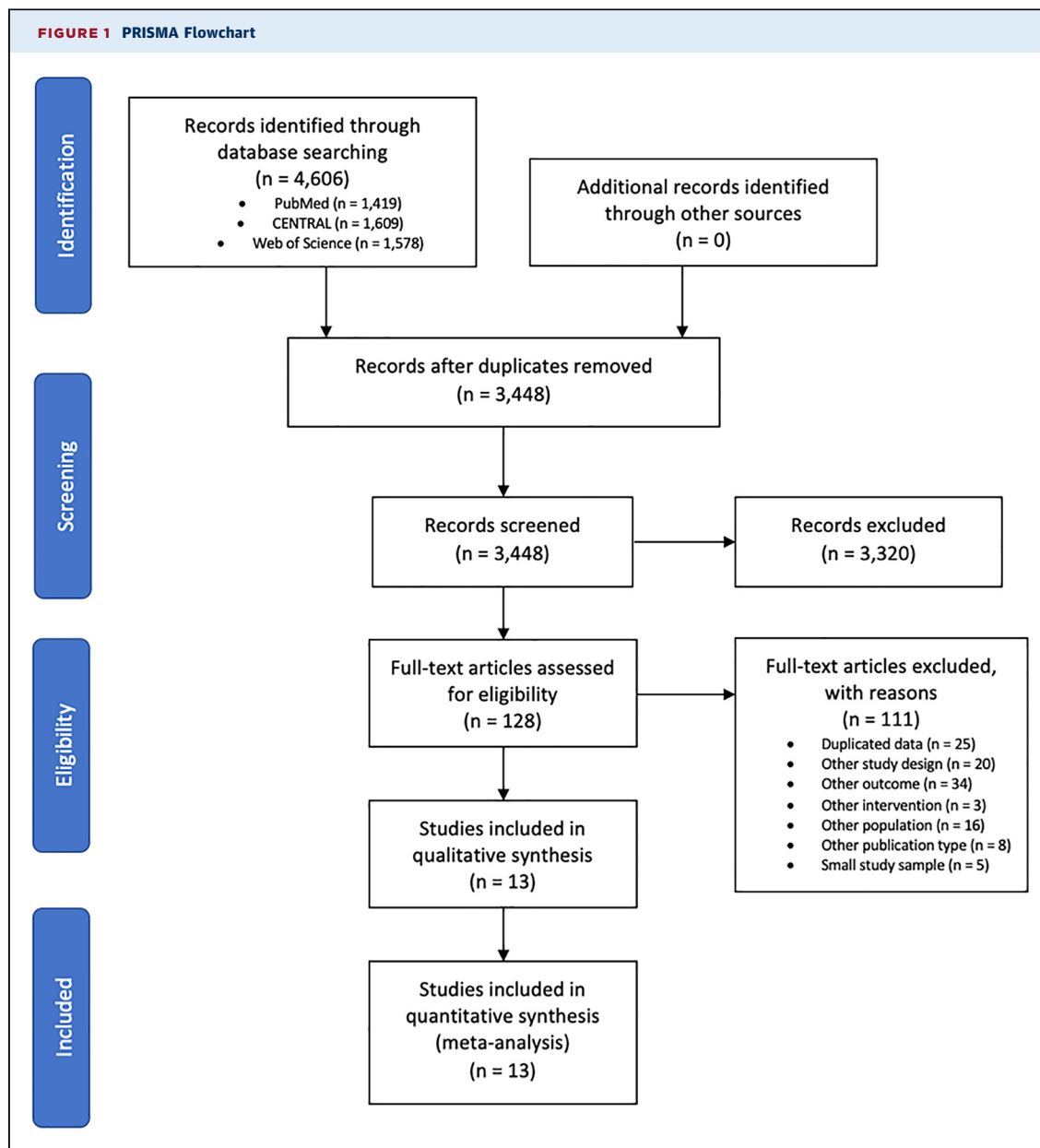
The network meta-analysis was performed using the "netmeta" package in R version 4.0.2 (R Foundation for Statistical Computing); values of $P < 0.05$ were considered statistically significant.

RESULTS

The search strategy yielded 3,488 studies; of those, 13 main studies and 10 substudies were deemed eligible.^{3-5,8,10-28} The PRISMA flow diagram is depicted in [Figure 1](#), and the pharmacological combinations networks across all outcomes are in [Supplemental Figure 1](#). A total of 29,875 patients (age 71.4 ± 9.0 years; 48.2% female) were analyzed with a mean follow-up of 33.0 months. Mean LVEF was $56.3\% \pm 8.7\%$, body mass index 29.9 ± 5.8 kg/m², NT-proBNP (N-terminal pro-B-type natriuretic peptide) 939.2 ± 812.1 pg/mL, and estimated glomerular filtration rate was 63.5 ± 20.0 mL/min. Two-thirds of patients were NYHA functional class II, and 38.6% had HF of ischemic etiology. Of all patients, 86.1% had a history of hypertension, 39.7% atrial fibrillation, and 39.5% diabetes ([Supplemental Table 1](#)). At baseline, excluding the trial intervention, 78.6% were treated with RASi, 3.4% with ARNI, 75.0% with BB, and 27.2% with MRA.

OUTCOMES. Nine studies with 27,978 patients reported the primary composite outcome of time to CV death or first HHF ([Supplemental Table 2](#)). SGLT2i (HR: 0.80 [95% CI: 0.74-0.88]), MRA (HR: 0.82 [95% CI: 0.69-0.98]), and ARNI (HR: 0.85 [95% CI: 0.72-0.99]) reduced the rates of the composite outcome, whereas RASi, BB, and digoxin did not ([Figure 2A](#)). The quadruple combinations of ARNI, BB, MRA, and SGLT2i (HR: 0.47 [95% CI: 0.31-0.70]; $P < 0.01$), and RASi, BB, MRA, and SGLT2i (HR: 0.51 [95% CI: 0.34-0.74]; $P < 0.01$), resulted in the most significant reductions ([Central Illustration A](#)). Notably, there was no significant difference between the combination of ARNI, BB, MRA, and SGLT2i and ARNI, MRA, and SGLT2i in reducing the primary outcome (HR: 0.84 [95% CI: 0.61-1.15]) ([Supplemental Table 3](#)).

Ten studies with 28,828 patients reported CV mortality. MRA reduced CV death (HR: 0.74 [95% CI: 0.57-0.97]) ([Figure 2B](#)). ARNI, BB, MRA, and SGLT2i (HR: 0.55 [95% CI: 0.32-0.95]; $P = 0.03$) and RASi, BB, MRA,

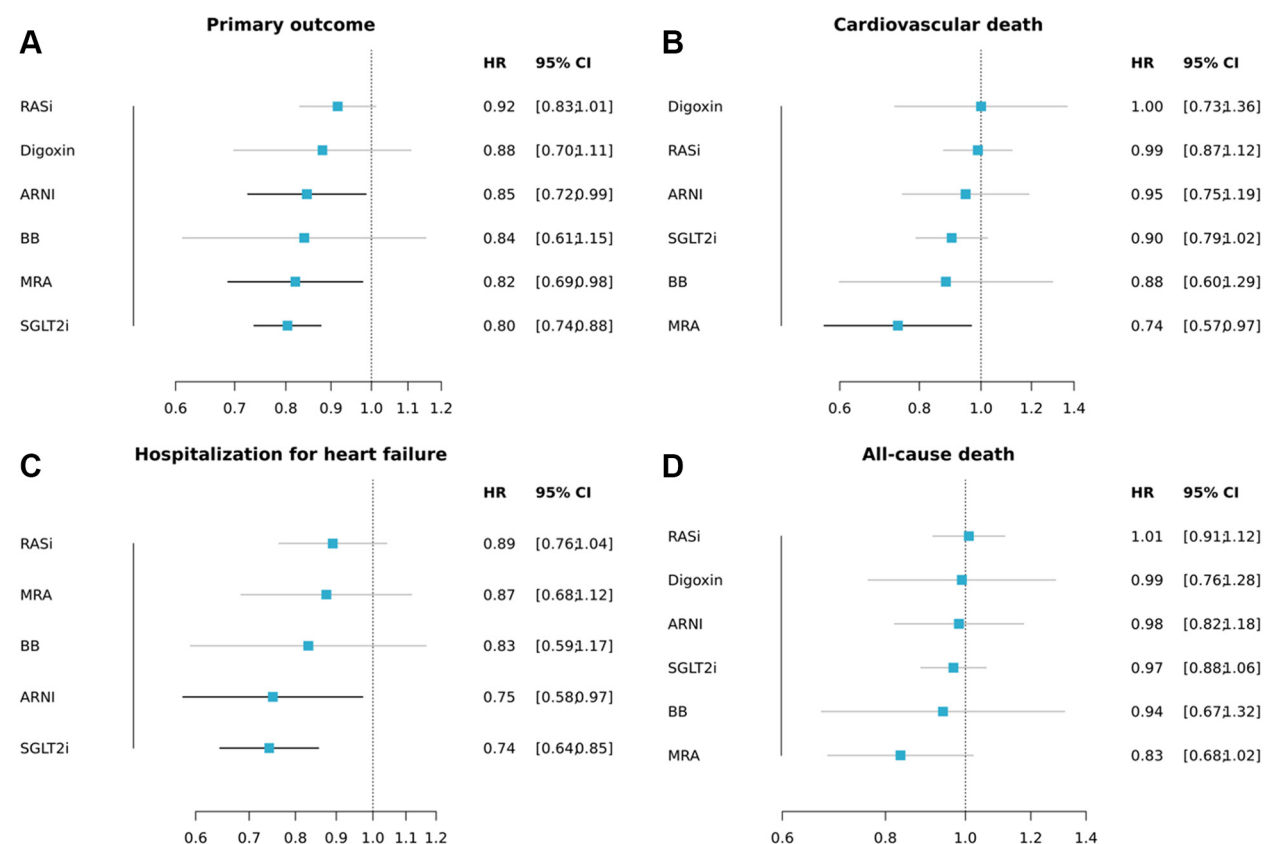


and SGLT2i (HR: 0.58 [95% CI: 0.35-0.96]; $P = 0.03$) combinations provided a survival benefit, mainly driven by the MRA effect (**Central Illustration B**). Like the primary outcome, there was no significant difference between the combination of ARNI, BB, MRA, and SGLT2i and ARNI, MRA, and SGLT2i in reducing CV death (HR: 0.88 [95% CI: 0.60-1.29]) (**Supplemental Table 4**).

Ten studies with 28,318 patients reported total HHF. SGLT2i (HR: 0.74 [95% CI: 0.64-0.85]) and ARNI

(HR: 0.75 [95% CI: 0.58-0.97]) reduced HHF (**Figure 2C**). A combination of ARNI, MRA, BB, and SGLT2i was the most effective in reducing total HHF (HR: 0.40 [95% CI: 0.24-0.67]; $P < 0.01$). Other results are shown in **Central Illustration C**. No difference was observed between the combination of ARNI, BB, MRA, and SGLT2i and ARNI, MRA, and SGLT2i in reducing total HHF (HR: 0.83 [95% CI: 0.59-1.17]) (**Supplemental Table 5**). Thirteen studies with 29,875 patients reported all-cause mortality; no individual

FIGURE 2 Forest Plots of Individual Drugs



Forest plots showing the HRs for (A) the primary outcome, (B) cardiovascular death (CV), (C) hospitalization for heart failure (HHF), and (D) all-cause death compared with placebo. ARNI = angiotensin receptor-neprilysin inhibitor; BB = beta-blocker; MRA = mineralocorticoid receptor antagonist; RASi = renin-angiotensin system inhibitor; SGLT2i = sodium glucose co-transporter 2 inhibitor.

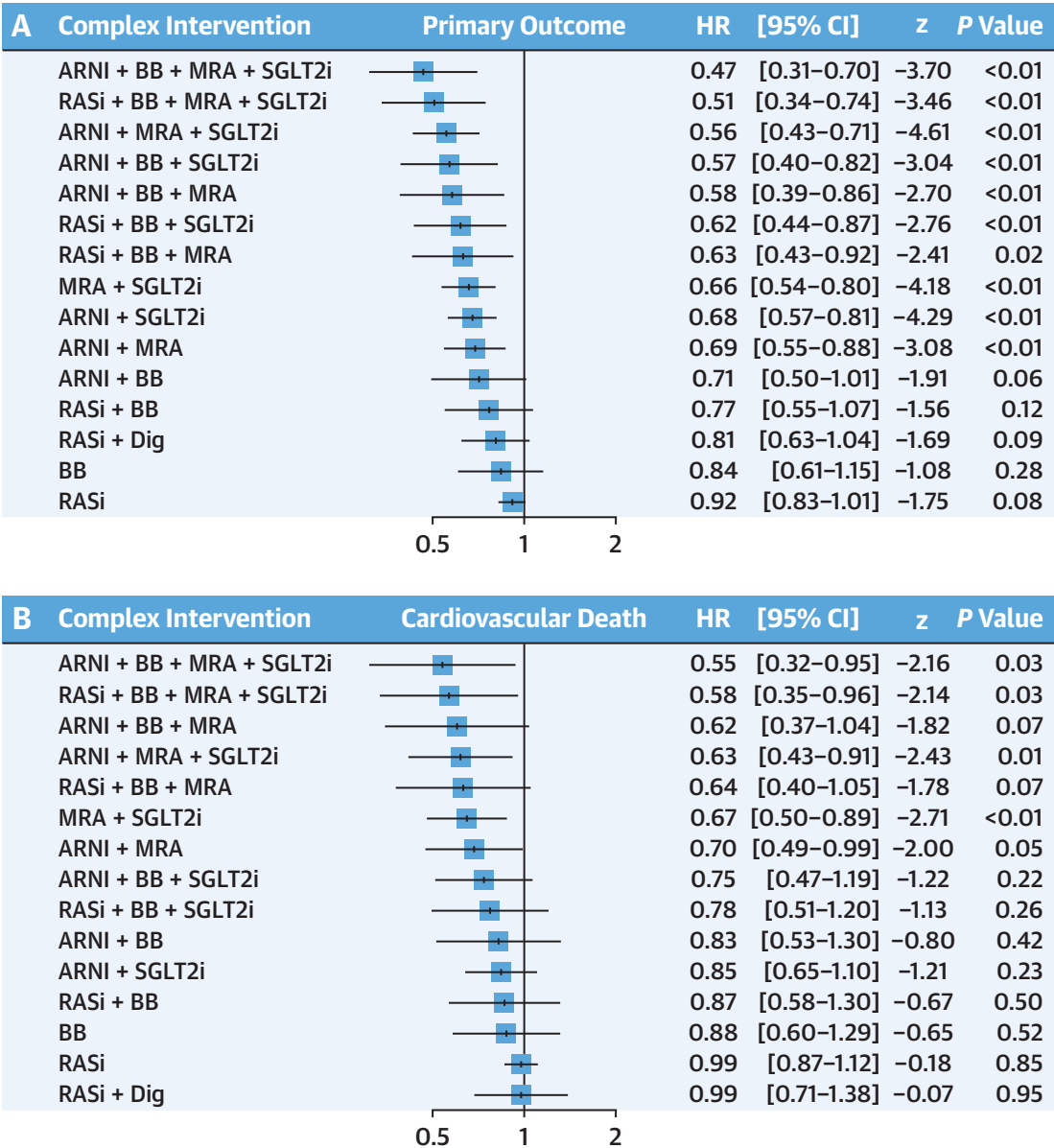
drug or combination resulted in risk reduction (Figure 2D, Central Illustration D, Supplemental Table 6).

The quadruple combination of ARNI, BB, MRA, and SGLT2i ranked first across all outcomes (P -score = 0.98 for the primary outcome, followed by RASi, BB, MRA, SGLT2i P -score = 0.88), ARNI, BB, SGLT2i (P -score = 0.74), and ARNI, BB, MRA (P -score = 0.70) (Supplemental Figure 2). The triple and all pairwise combinations of ARNI, MRA, and SGLT2i demonstrated statistically robust improvements for the primary endpoint and total HHF (Supplemental Figures 3A and 3C); only combinations that included MRA reduced CV death (Supplemental Figure 3B) but none reduced all-cause death (Supplemental Figure 3D).

SUBGROUP ANALYSIS ACCORDING TO LVEF. A subgroup analysis of patients with HFmrEF and

HFpEF was performed. ARNI, MRA, or SGLT2i separately reduced the primary outcome in patients with HFmrEF whereas only SGLT2i improved the primary outcome in HFpEF patients (Figure 3A). The triple combination of ARNI, MRA, and SGLT2i was the most effective in reducing the rates of the primary outcome among HFmrEF patients (HR: 0.28 [95% CI: 0.15-0.54]), whereas in patients with HFpEF the same combination yielded an attenuated reduction (HR: 0.62 [95% CI: 0.42-0.93]) (Figure 3A). Additionally, the triple combination reduced both CV death and total HHF only in HFmrEF but not in HFpEF patients (Figures 3B and 3C). A further subgroup analysis of HFpEF patients with a cutoff of LVEF 60% subgroups using data from PARAGON-HF (Prospective Comparison of ARNI with ARB Global Outcomes in HF With Preserved Ejection Fraction), TOPCAT, EMPEROR-Preserved (Empagliflozin Outcome Trial in Patients

CENTRAL ILLUSTRATION Combination Treatment Effects



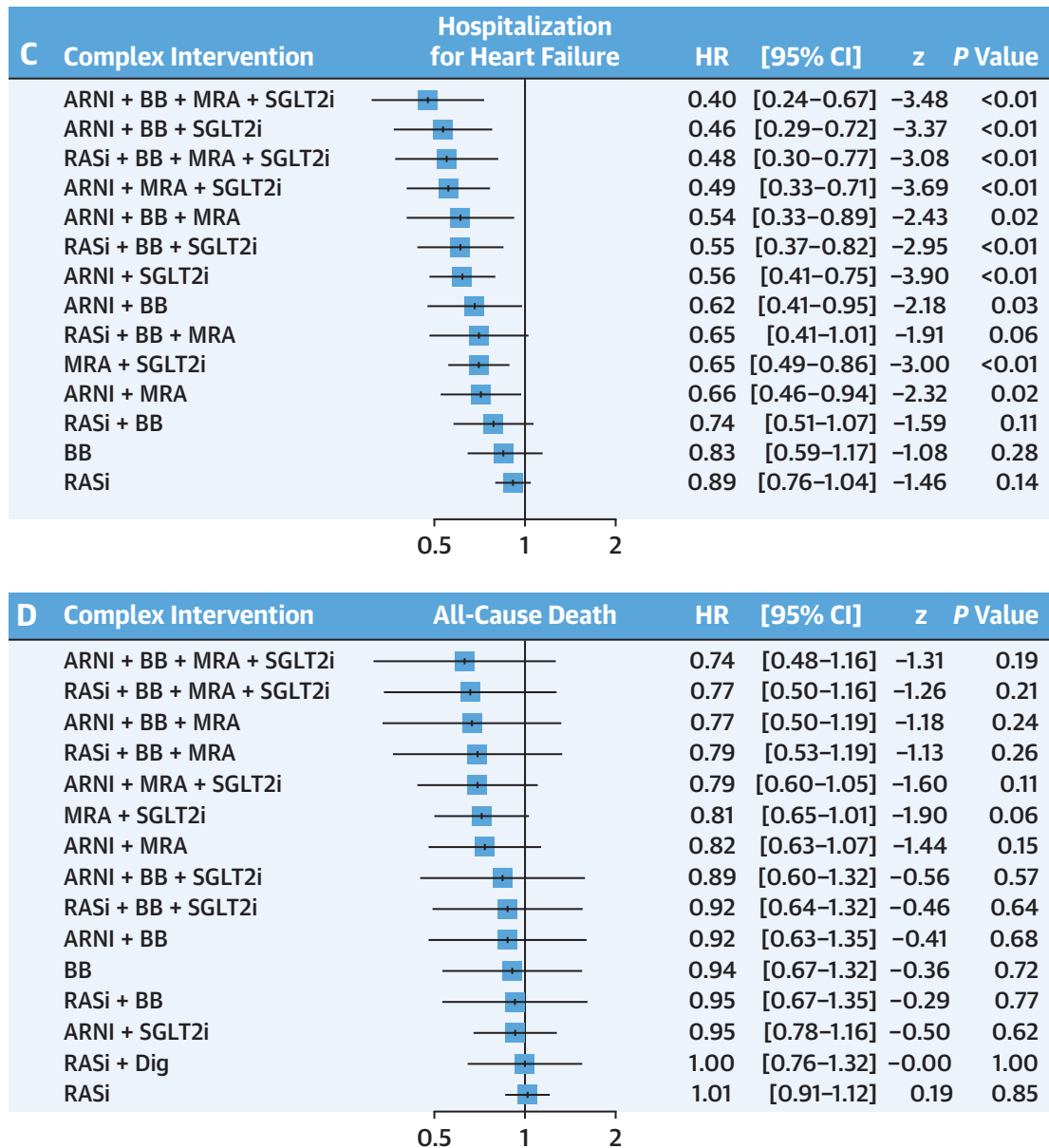
Zafeiropoulos S, et al. J Am Coll Cardiol HF. 2024;12(4):616–627.

Combination of treatment effect on (A) the primary outcome, (B) cardiovascular death, (C) heart failure hospitalizations, and (D) all-cause death compared with null therapy. ARNI = angiotensin receptor-neprilysin inhibitor; BB = beta-blocker; Dig = digoxin; MRA = mineralocorticoid receptor antagonist; RASi = renin-angiotensin system inhibitor; SGLT2i = sodium glucose co-transporter 2 inhibitor.

Continued on the next page

With Chronic Heart Failure With Preserved Ejection Fraction), and DELIVER (Dapagliflozin Evaluation to Improve the LIVES of Patients With PReserved Ejection Fraction Heart Failure) trials was performed. Compared with a background therapy of RASi, the

triple combination of ARNI, MRA, and SGLT2i yielded a substantial reduction (HR: 0.62 [95% CI: 0.45–0.85]) in the LVEF 50% to 60% subgroup and a marginally significant benefit (HR: 0.72 [95% CI: 0.52–0.99]) among patients with LVEF $\geq$ 60% (Figure 4A).

CENTRAL ILLUSTRATION Continued

Zafeiropoulos S, et al. J Am Coll Cardiol HF. 2024;12(4):616–627.

Moreover, the triple combination did not affect CV death in both subgroups (Figure 4B), whereas it reduced total HHF in the LVEF 50% to 59% but not $\geq 60\%$ subgroup (Figure 4C).

QUALITY ASSESSMENT. Risk of bias and within-study bias was low (Supplemental Table 7, Supplemental Figure 4). There was no evidence of systematic reporting bias; the composite of CV mortality and HF hospitalization (Egger's test $P = 0.87$), CV death alone (Egger's test $P = 0.59$), HHF alone

(Egger's test $P = 0.27$), and all-cause mortality (Egger's test $P = 0.44$) (Supplemental Figure 5). Indirectness was judged as low, and most comparisons did not show concerns of imprecision. The confidence rating of most comparisons was considered moderate (Supplemental Table 8).

SENSITIVITY ANALYSIS. In the sensitivity analysis with the full cohort of TOPCAT, the benefit of MRA as an individual drug was blunted (Supplemental

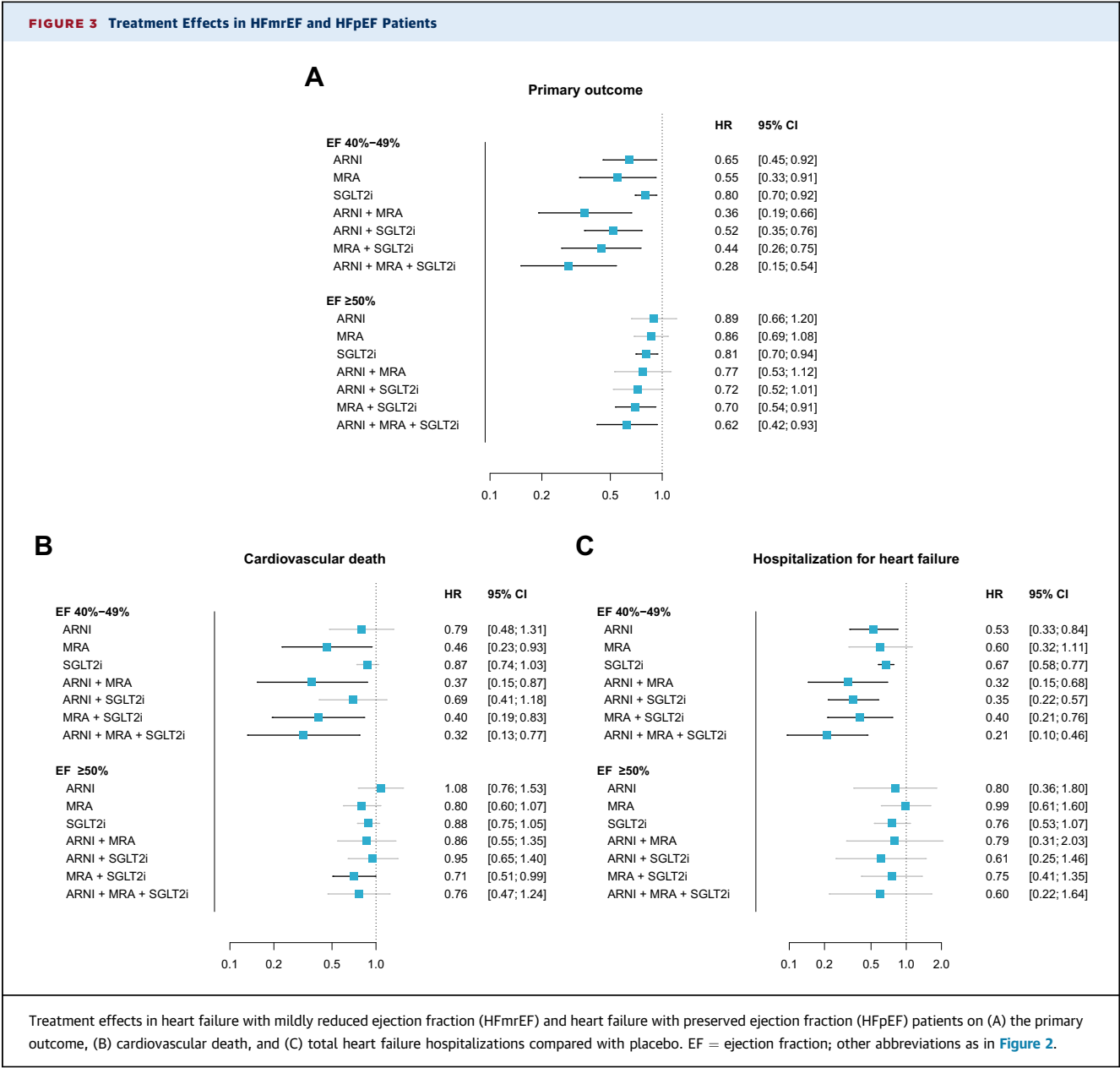


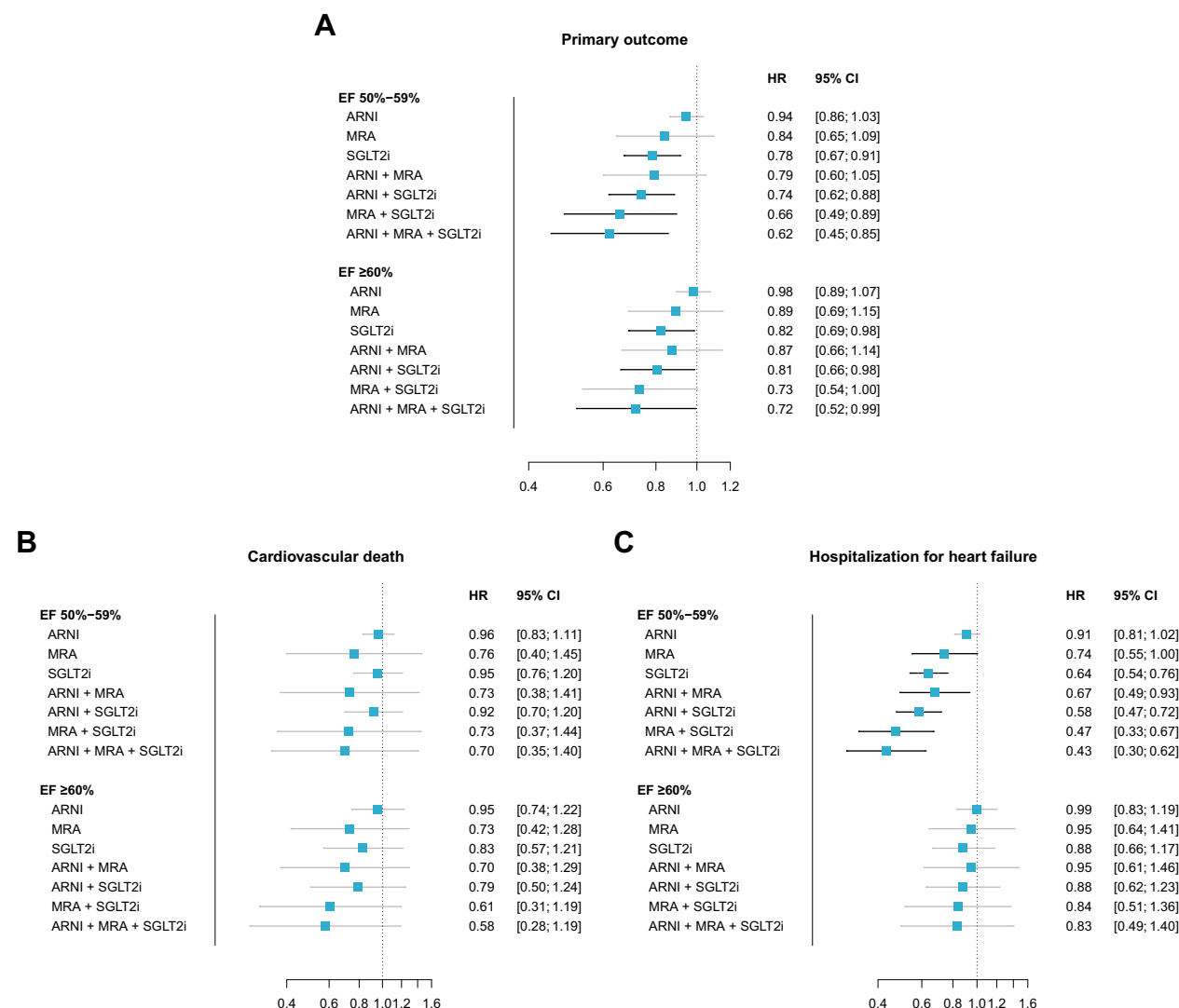
Figure 6). The effect of its combinations was also partially attenuated; however, its combination with ARNI, BB, and SGLT2i remained effective in improving the primary composite outcome and HHF, but not CV and all-cause death (Supplemental Figure 7).

DISCUSSION

In this additive component network meta-analysis, we demonstrated that a pharmacological combination of ARNI, BB, MRA, and SGLT2i was the most effective in reducing a composite of time to CV death or first HHF, as well as the CV death and total HHF in

HFmrEF/HFpEF patients. This benefit was largely attributed to the effect of the triple combination of ARNI, MRA, and SGLT2i, because there was minimal incremental benefit when BB were added. Notably, the triple combination benefit was more pronounced in the HFmrEF population and was extended, but gradually attenuated, in patients with HF and LVEF 50% to 59% and ≥60%; although SGLT2i were consistently effective across all LVEF subgroups (LVEF 40% to 49%, 50% to 59%, ≥60%). Recently, a standard network meta-analysis showed that SGLT2i was the most beneficial drug and pointed toward incremental benefits in HHF—but

FIGURE 4 Treatment Effects in HFpEF Patients Using an LVEF Cutoff of 60%



Treatment effects in patients with a left ventricular ejection fraction (LVEF) 50% to 59%, and $\geq 60\%$ on (A) the primary outcome, (B) cardiovascular death, and (C) total heart failure hospitalizations compared with background therapy of RASi. Abbreviations as in [Figure 2](#).

not all-cause death—with a combination of RASi, BB, and SGLT2i. However, they were not able to explore many treatment combinations or CV death.²⁹ To our knowledge, this is the first additive component network meta-analysis that provides the most comprehensive analysis of pharmacological combinations' effects across multiple clinical outcomes in patients with HF and LVEF $>40\%$. Our findings support the concept of treating these patients with a triple combination of ARNI, MRA, and SGLT2i.

In HFmrEF patients, the combination of ARNI, MRA, and SGLT2i had the greatest reduction in the

risk of a composite of CV death and HHF, as well as CV death and total HHF alone. The latest guidelines by the European Society of Cardiology and AHA/ACC/HFSA provide a Class 2a recommendation for SGLT2i—although new evidence has emerged since their publication—and Class 2b for ARNI and MRA for patients with HFmrEF. Our findings show a substantial benefit of this triple combination and support its usage in this population.

In patients with HFpEF, the same triple combination was effective in reducing the composite primary outcome without reaching statistical significance for CV death or total HHF. Notably, in LVEF 50% to 59%

stratum, this combination resulted in a robust reduction in the primary outcome and total HHF, but not CV death. Additionally, in the LVEF $\geq 60\%$ stratum, this triple combination marginally improved the primary outcome without reaching statistical significance for either CV death or total HHF. Overall, our findings show that a meaningful, yet attenuated, benefit of this triple combination may extend to the HFpEF population, particularly in the LVEF 50% to 59% subgroup.

Recently, a new and simplified HF classification was proposed.³⁰ It classifies patients into those with reduced LVEF and those with “normal” LVEF with the cutoff of LVEF around 55% to 60%, based on demographic and clinical factors, and suggests the use of GDMT in the first class. In line with this suggestion, our analysis shows a remarkable benefit of the triple combination of ARNI, MRA, and SGLT2i in HF patients with LVEF up to 60%, whereas there was a blunted benefit in HF patients and LVEF $\geq 60\%$. Therefore, a broader indication of neurohormonal treatments in HF patients with LVEF up to 55% to 60% might be reasonable, whereas SGLT2i has been shown to improve CV outcomes irrespective of LVEF.³¹

BB have no large, adequately powered RCTs in patients exclusively with HF and LVEF $>40\%$, and the best available estimates are based on an individual patient-level meta-analysis of substudies.³² In our analysis, given the fact that three-fourths of our total population was already treated with BB, possibly due to comorbidities (eg, atrial fibrillation, coronary artery disease), BB were part of the beneficial quadruple combination, but we found no evidence of an additional benefit on top of the triple combination of ARNI, MRA, and SGLT2i in the overall population. Notwithstanding, the benefits of the GDMT of HFmrEF might extend to the majority of HFmrEF patients.³² However, lately, there is emerging evidence that low heart rate might worsen outcomes in patients with HFpEF,³³ and thus these patients might benefit from BB withdrawal.^{34,35} Our analysis does not exclude the possibility of a modest beneficial contribution of BB in the combination treatment, but more research is warranted regarding the role of BB in patients with HF and LVEF $>40\%$. Similarly, digoxin did not improve any cardiovascular outcome in our analysis, and observational studies suggest that it is associated with increased morbidity and mortality in older HFpEF patients.³⁶

A recent analysis of data from PARAGON-HF, TOPCAT, and EMPEROR-Preserved showed that a

combination of ARNI, MRA, and SGLT2i provides a benefit that extends up to LVEF 65%, although CV death was reduced only in the LVEF 45% to 55% subgroup.³⁷ Beyond the different methodologies, we expand on that analysis by: 1) including estimates from 13 randomized controlled clinical trials; 2) including more pharmacological treatments, that is, RASi, BB, digoxin; and 3) including additional clinical outcomes, that is, all-cause mortality. Interestingly, the recent STRONG-HF (Safety, Tolerability and Efficacy of Rapid Optimization, Helped by NT-proBNP testing, of Heart Failure Therapies) trial showed that an intensive treatment strategy of rapid up-titration of RASi, BB, and MRA after an HHF, improved clinical outcomes compared with the standard of care.³⁸ Notably, this study enrolled patients regardless of their ejection fraction, and the intervention was equally effective in HFmrEF, HFpEF, and HFpEF patients, suggesting a beneficial effect of this combination in the early post-discharge period regardless of LVEF.³⁸ The addition of SGLT2i in this strategy, which has the best evidence of efficacy across the LVEF spectrum, may be reasonable.³¹

STUDY LIMITATIONS. We did not account for different drug dosages. The 50% threshold to define the use of each drug class on the background therapy may be a limitation although it is commonly used in similar analyses.⁷ Additionally, we retrieved all the published substudies related to the parent trials to extract data from subgroups that were treated entirely with the eligible drug classes and thus achieving the maximum homogeneity and accuracy in the treatment combination assignment. Although ARNI, MRA, and SGLT2i target distinct pathways, there might be an overlap between their effects. No significant treatment interactions between ARNI and other drugs have been reported in individual trials, and as such, none was inserted in our model.³ An interaction between MRA and empagliflozin in HHF outcome was reported,²⁵ but not been confirmed, in a subsequent trial.²³ Due to the known enrollment and adherence issues in the TOPCAT,²⁰ we used data from the Americas cohort for primary analysis and performed a sensitivity analysis with the full TOPCAT trial, a strategy that has been implemented previously.³⁷ Lastly, the DELIVER trial had a slightly different definition of the primary outcome, which also included urgent visits for HF; however, this did not lead to any substantial change in the HR estimates.³¹

CONCLUSIONS

This additive network meta-analysis suggests that a quadruple combination of ARNI, BB, MRA, and SGLT2i had the greatest estimated aggregate benefit in HF patients with LVEF >40%. This benefit is largely driven by the robust and consistent effect of the triple combination of ARNI, MRA, and SGLT2i, which is more pronounced in HFmrEF patients, remains robust in LVEF 50% to 59% subgroup, and is attenuated, yet significant, in HF patients with LVEF ≥60%.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Konstantinides is a consultant to Daiichi-Sankyo, Bayer AG, Boston Scientific, Penumbra Inc, and LumiraDx; and he has received honoraria for lectures from Bristol-Myers Squibb-Pfizer, Boston Scientific, and MSD. Dr Giamouzis has received honoraria for lectures and/or advisory boards from AstraZeneca, Bayer, Boehringer Ingelheim, ELPEN Pharmaceuticals, Genesis Pharma, Menarini Hellas, Novartis, Pfizer, Roche, and Servier. Dr Cooper is research consultant for Abiomed. Dr Butler is a consultant to Abbott, Adrenomed, American Regent, Arca Biopharma, Amgen, Applied Therapeutic, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Cardiac Dimension, Cardior CVRx, Cytokinetics, Edwards Lifesciences, Element Science, Fast Biomedical, G3 Pharmaceutical, Innolife, Impulse Dynamics, Imbria, Inventiva, Lexicon, Lilly, LivaNova Janssen, Johnson & Johnson, Medtronic, Merck, Occlutech, Nestle, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Roche, Sanofi, Sequana, SQ Innovation, 3live, Velakor, and Vifor. Dr Giannakoulas has received honoraria for lectures and/or advisory boards from AstraZeneca, Boehringer Ingelheim, Novartis, Servier, Roche, Pfizer, ELPEN Pharmaceuticals, and Genesis Pharma. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr George Giannakoulas, Cardiology Department, AHEPA Hospital, Aristotle University of Thessaloniki, Stilp. Kiriakidi 1, 54637 Thessaloniki, Greece. E-mail: ggiannakoulas@auth.gr.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: This additive network meta-analysis demonstrates that combination treatments in patients with HF and LVEF >40% could reduce the risk of a combined outcome of CV death and first HHF as well as CV death and total HHF alone. Specifically, it highlights the combination of ARNI, MRA and SGLT2i as the most robust and consistent. However, the triple combination effect is more pronounced in HF patients on the lower end of this LVEF spectrum and is attenuated on the upper end.

TRANSLATIONAL OUTLOOK: In HFmrEF/HFpEF patients, the quadruple combination of ARNI, BB, MRA and SGLT2i is the most effective but this is largely explained by the effect of the triple combination of ARNI, MRA and SGLT2i. Therefore, serious efforts should be made to treat these patients with comprehensive pharmacological therapy. However, the role of BB in this population requires further research.

REFERENCES

- Pfeffer MA, Shah AM, Borlaug BA. Heart failure with preserved ejection fraction in perspective. *Circ Res*. 2019;124(11):1598-1617.
- Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2022;79(17):e263-e421. <https://doi.org/10.1016/j.jacc.2021.12.012>
- Anker SD, Butler J, Filippatos G, et al. Empagliflozin in heart failure with a preserved ejection fraction. *N Engl J Med*. 2021;385(16):1451-1461. <https://doi.org/10.1056/NEJMoa2107038>
- Solomon SD, McMurray JJV, Claggett B, et al. Dapagliflozin in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med*. 2022;387(12):1089-1098.
- Solomon SD, McMurray JJV, Anand IS, et al. Angiotensin-neprilysin inhibition in heart failure with preserved ejection fraction. *N Engl J Med*. 2019;381(17):1609-1620.
- Hutton B, Salanti G, Caldwell DM, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Annals of Internal Medicine*. 2015;162(11):777-784.
- Tromp J, Ouwerkerk W, Veldhuisen DJ, et al. A systematic review and network meta-analysis of pharmacological treatment of heart failure with reduced ejection fraction. *J Am Coll Cardiol HF*. 2022;10(2):73-84.
- Nikolakopoulou A, Higgins JPT, Papakonstantinou T, et al. CINeMA: an approach for assessing confidence in the results of a network meta-analysis. *PLOS Med*. 2020;17:e1003082.
- Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898.
- Solomon SD, Zile M, Pieske B, et al. The angiotensin receptor neprilysin inhibitor LCZ696 in heart failure with preserved ejection fraction: a phase 2 double-blind randomised controlled trial. *Lancet*. 2012;380(9851):1387-1395.
- Pitt B, Pfeffer MA, Assmann SF, et al. Spironolactone for heart failure with preserved ejection fraction. *N Engl J Med*. 2014;370(15):1383-1392.
- Massie BM, Carson PE, McMurray JJ, et al. Irbesartan in patients with heart failure and preserved ejection fraction. *N Engl J Med*. 2008;359(23):2456-2467.
- Yusuf S, Pfeffer MA, Swedberg K, et al. Effects of candesartan in patients with chronic heart failure and preserved left-ventricular ejection fraction: the CHARM-Preserved trial. *Lancet*. 2003;362(9386):777-781.
- Edelmann F, Wachter R, Schmidt AG, et al. Effect of spironolactone on diastolic function and exercise capacity in patients with heart failure with preserved ejection fraction: the Aldo-DHF randomized controlled trial. *JAMA*. 2013;309(8):781-791.
- Cleland JGF, Tendera M, Adamus J, Freemantle N, Polonski L, Taylor J. The perindopril in elderly people with chronic heart failure (PEP-CHF) study. *Eur Heart J*. 2006;27(19):2338-2345.

16. van Veldhuisen DJ, Cohen-Solal A, Böhm M, et al. Beta-blockade with nebivolol in elderly heart failure patients with impaired and preserved left ventricular ejection fraction: data from SENIORS (Study of Effects of Nebivolol Intervention on Outcomes and Rehospitalization in Seniors With Heart Failure). *J Am Coll Cardiol*. 2009;53(23):2150–2158.
17. Ahmed A, Rich MW, Love TE, et al. Digoxin and reduction in mortality and hospitalization in heart failure: a comprehensive post hoc analysis of the DIG trial. *Eur Heart J*. 2006;27(2):178–186.
18. Nassif ME, Windsor SL, Borlaug BA, et al. The SGLT2 inhibitor dapagliflozin in heart failure with preserved ejection fraction: a multicenter randomized trial. *Nat Med*. 2021;27(11):1954–1960.
19. Anker SD, Butler J, Usman MS, et al. Efficacy of empagliflozin in heart failure with preserved versus mid-range ejection fraction: a pre-specified analysis of EMPEROR-Preserved. *Nat Med*. 2022;28(12):2512–2520.
20. Pfeffer MA, Claggett B, Assmann SF, et al. Regional variation in patients and outcomes in the Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist (TOPCAT) trial. *Circulation*. 2015;131(1):34–42.
21. Solomon SD, Claggett B, Lewis EF, et al. Influence of ejection fraction on outcomes and efficacy of spironolactone in patients with heart failure with preserved ejection fraction. *Eur Heart J*. 2016;37(5):455–462.
22. Jhund PS, Kondo T, Butt JH, et al. Dapagliflozin across the range of ejection fraction in patients with heart failure: a patient-level, pooled meta-analysis of DAPA-HF and DELIVER. *Nat Med*. 2022;28(9):1956–1964.
23. Yang M, Butt JH, Kondo T, et al. Dapagliflozin in patients with heart failure with mildly reduced and preserved ejection fraction treated with a mineralocorticoid receptor antagonist or sacubitril/valsartan. *Eur J Heart Fail*. 2022;24(12):2307–2319.
24. Butler J, Packer M, Filippatos G, et al. Effect of empagliflozin in patients with heart failure across the spectrum of left ventricular ejection fraction. *Eur Heart J*. 2022;43(5):416–426.
25. Ferreira JP, Butler J, Zannad F, et al. Mineralocorticoid receptor antagonists and empagliflozin in patients with heart failure and preserved ejection fraction. *J Am Coll Cardiol*. 2022;79(12):1129–1137.
26. Jering KS, Zannad F, Claggett B, et al. Cardiovascular and Renal Outcomes of Mineralocorticoid Receptor Antagonist Use in PARAGON-HF. *J Am Coll Cardiol HF*. 2021;9(1):13–24.
27. Solomon SD, Vaduganathan M, Claggett BL, et al. Sacubitril/valsartan across the spectrum of ejection fraction in heart failure. *Circulation*. 2020;141(5):352–361.
28. Lund LH, Claggett B, Liu J, et al. Heart failure with mid-range ejection fraction in CHARM: characteristics, outcomes and effect of candesartan across the entire ejection fraction spectrum. *Eur J Heart Fail*. 2018;20(8):1230–1239.
29. Xiang B, Zhang R, Wu X, Zhou X. Optimal pharmacologic treatment of heart failure with preserved and mildly reduced ejection fraction: a meta-analysis. *JAMA Netw Open*. 2022;5(9):e2231963–e2231963.
30. Docherty KF, Bayes-Genis A, Butler J, et al. The four pillars of HFmrEF therapy: is it time to treat heart failure regardless of ejection fraction? *Eur Heart J Suppl*. 2022;24(supplement_L):L10–L19.
31. Vaduganathan M, Docherty KF, Claggett BL, et al. SGLT-2 inhibitors in patients with heart failure: a comprehensive meta-analysis of five randomized controlled trials. *Lancet*. 2022;400(10354):757–767.
32. Cleland JGF, Bunting KV, Flather MD, et al. Beta-blockers for heart failure with reduced, mid-range, and preserved ejection fraction: an individual patient-level analysis of double-blind randomized trials. *Eur Heart J*. 2018;39(1):26–35.
33. Infeld M, Wahlberg K, Cicero J, et al. Effect of personalized accelerated pacing on quality of life, physical activity, and atrial fibrillation in patients with preclinical and overt heart failure with preserved ejection fraction: the myPACE randomized clinical trial. *JAMA Cardiol*. 2023;8(3):213–221. <https://doi.org/10.1001/jamacardio.2022.5320>
34. Palau P, Seller J, Domínguez E, et al. Effect of β -blocker withdrawal on functional capacity in heart failure and preserved ejection fraction. *J Am Coll Cardiol*. 2021;78(21):2042–2056.
35. Silverman DN, Plante TB, Infeld M, et al. Association of β -blocker use with heart failure hospitalizations and cardiovascular disease mortality among patients with heart failure with a preserved ejection fraction: a secondary analysis of the TOPCAT trial. *JAMA Netw Open*. 2019;2(12):e1916598–e1916598.
36. Llàcer P, Núñez J, Bayés-Genis A, et al. Digoxin and prognosis of heart failure in older patients with preserved ejection fraction: importance of heart rate. Results from an observational and multicenter study. *Eur J Intern Med*. 2019;60:18–23.
37. Vaduganathan M, Claggett BL, Inciardi RM, Fonarow GC, McMurray JJV, Solomon SD. Estimating the benefits of combination medical therapy in heart failure with mildly reduced and preserved ejection fraction. *Circulation*. 2022;145(23):1741–1743.
38. Mebazaa A, Davison B, Chioncel O, et al. Safety, tolerability and efficacy of up-titration of guideline-directed medical therapies for acute heart failure (STRONG-HF): a multinational, open-label, randomised, trial. *Lancet*. 2022;400(10367):1938–1952. [https://doi.org/10.1016/S0140-6736\(22\)02076-1](https://doi.org/10.1016/S0140-6736(22)02076-1)

KEY WORDS angiotensin receptor-neprilysin inhibitor, heart failure with preserved ejection fraction, mineralocorticoid receptor antagonists, network meta-analysis, sodium glucose co-transporter 2 inhibitors

APPENDIX For the search string and supplemental figures and tables, please see the online version of this paper.